

Questions

1. Beginner

#1. List all books in the Fiction genre.

```
SELECT * FROM books WHERE genre = 'Fiction';
```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
book_id	title	author	genre	price
19	Sisters by Choice	Susan Mallery	Fiction	14.99
20	Charged	G. P. Ching	Fiction	4.99
21	Driving Mr. Wrong Home	Shirley Jump	Fiction	1.99
22	Familiar Faces	Mary Roberts Rinehart	Fiction	9.99
23	Cocoa Beans	Michael J. Hughes	Fiction	3.99
25	In the Dead of the Night	Terry Spear	Fiction	4.99
26	Collected Fiction	Irwin Shaw	Fiction	29.99
27	The Madame Catches Her Duke	Christina McKnight	Fiction	4.99
29	Echoes of Chaos: An Epic Fantasy Novel (The F...	J.D.L. Rosell	Fiction	4.99
31	Adhd Isnt Me!	Candace Curry	Fiction	5.99
32	Trapped in Oz	Andrew Grev	Fiction	4.99

#2. Find all the books priced above 500.

```
SELECT * FROM books WHERE price > 500;
```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
book_id	title	author	genre	price
196	Machine Learning: Concepts, Methodologies, To...	Management Association	Computers	1755.00
246	The SAGE International Encyclopedia of Travel ...	Linda L. Lowry	Business & Economics	548.00

#3. Identify the top 10 most expensive books.

```
SELECT * FROM books ORDER BY price DESC LIMIT 10;
```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:	Fetch rows:
book_id	title	author	genre	price	
196	Machine Learning: Concepts, Methodologies, To...	Management Association	Computers	1755.00	
246	The SAGE International Encyclopedia of Travel ...	Linda L. Lowry	Business & Economics	548.00	
43	Encyclopedia of Time	Samuel L. Macey	Business & Economics	300.00	
224	Skin Cancer: A Comprehensive Guide	Keyvan Nouri	Medical	290.00	
471	Italian Literature since 1900 in English Translation	Robin Healey	Literary Criticism	256.00	
220	Machine Learning for Societal Improvement, Mo...	Pendyala	Computers	255.00	
226	Biomedical and Business Applications Using Arti...	Segall	Computers	255.00	
327	Formulating Pharma-, Nutra-, and Cosmeceutic...	Anupama Singh	Medical	239.00	
267	An Introductory Philosophy of Medicine	James A. Marcum	Medical	239.00	
126	History of Technology Volume 10	Norman Smith	History	234.00	

#4. List all the delivered orders.

```
SELECT * FROM orders WHERE order_status = 'Delivered';
```

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

	order_id	member_id	book_id	order_date	quantity	order_status
▶	2	44	105	2024-12-17	3	Delivered
	3	65	251	2024-05-08	1	Delivered
	4	55	425	2024-05-21	1	Delivered
	5	43	395	2024-03-07	3	Delivered
	7	55	288	2024-01-05	1	Delivered
	9	71	76	2024-08-08	1	Delivered
	10	40	187	2024-01-21	2	Delivered
	12	53	499	2024-11-13	3	Delivered
	13	31	443	2024-03-24	1	Delivered
	15	32	137	2024-03-22	3	Delivered
	16	49	447	2024-01-20	2	Delivered
	17	26	419	2024-08-23	2	Delivered

#5. Show all premium members

```
SELECT * FROM members WHERE membership_type = 'Premium';
```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
member_id	name	city	join_date	membership_type
1	Nikhil Menon	Kochi	2024-01-30	Premium
3	Sneha Jose	Kochi	2023-08-28	Premium
6	Aravind Jose	Thrissur	2022-08-14	Premium
8	Jishnu Pillai	Trivandrum	2023-03-09	Premium
9	Anu Pillai	Ernakulam	2024-02-21	Premium
10	Rahul Menon	Thrissur	2022-04-10	Premium
12	Devika Iyer	Kozhikode	2022-05-08	Premium
13	Meera Jose	Kottayam	2024-04-29	Premium
16	Rekha Kumar	Alappuzha	2022-04-14	Premium
17	Anu Iyer	Trivandrum	2024-05-04	Premium
18	Sanjay Nam...	Kottayam	2022-08-03	Premium
20	Manoj Kumar	Ernakulam	2023-04-19	Premium
21	Anu Jose	Ernakulam	2023-12-03	Premium

#6. List all cancelled orders

```
SELECT * FROM orders WHERE order_status = 'Cancelled';
```

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

	order_id	member_id	book_id	order_date	quantity	order_status
	46	18	411	2024-08-25	3	Cancelled
	47	72	305	2024-06-11	2	Cancelled
	55	9	213	2024-07-27	2	Cancelled
	59	39	386	2024-07-18	2	Cancelled
	63	22	431	2024-08-27	1	Cancelled
	64	69	14	2024-07-20	3	Cancelled
	74	33	393	2024-07-07	1	Cancelled
	76	14	297	2024-01-14	2	Cancelled
	77	49	204	2024-04-11	1	Cancelled

#7. Find all books by a specific author.

```
SELECT * FROM books WHERE author = 'Ivan King';
```

Result Grid					
Filter Rows:					
Edit:   					
Export/Import:  					
Wrap Cell Content: 					
	book_id	title	author	genre	price
▶	1	Bestsellers	Ivan King	Young Adult Fiction	12.99
	5	Motivational Books	Ivan King	Self-Help	12.99
	6	8 Great Habits of Extremely Effective Christians	Ivan King	Self-Help	12.99
	7	Purpose of a Christian Life	Ivan King	Religion	12.99
	8	Self Help Books	Ivan King	Self-Help	12.99
	9	Personal Development	Ivan King	Business & Economics	9.99
	10	Inspirational Christian	Ivan King	Young Adult Fiction	9.99
	11	Christian Books	Ivan King	Religion	9.99
	12	Books About Life	Ivan King	Science	9.99

2. Intermediate

#8. Calculate total revenue generated per genre.

```
SELECT b.genre, SUM(b.price * o.quantity) AS revenue
FROM orders o
JOIN books b ON o.book_id = b.book_id
GROUP BY b.genre
ORDER BY revenue DESC;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
	genre	revenue	
▶	Computers	19016.15	
	Business & Economics	13110.13	
	Literary Criticism	7429.62	
	Language Arts & Disciplines	6377.72	
	Philosophy	5543.05	
	History	5087.80	
	Psychology	3780.10	
	Technology & Engineering	3269.94	
	Social Science	3194.48	

#9. Find the top 5 members by total spending.

```
SELECT m.name, SUM(b.price * o.quantity) AS total_spent
FROM orders o
JOIN members m ON o.member_id = m.member_id
JOIN books b ON o.book_id = b.book_id
GROUP BY m.name
ORDER BY total_spent DESC
LIMIT 5;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
	name	total_spent		
▶	Sanjay Nambiar	5858.51		
	Ajay Menon	3900.70		
	Arun Nair	3846.31		
	Lakshmi Jose	3679.77		
	Manoj Kumar	2639.17		

#10. Calculate the average order value per membership type.

```
SELECT m.membership_type, AVG(b.price * o.quantity) AS avg_order_value
FROM orders o
JOIN members m ON o.member_id = m.member_id
JOIN books b ON o.book_id = b.book_id
GROUP BY m.membership_type;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
	membership_type	avg_order_value	
▶	Premium	117.113271	
	Regular	100.248703	

#11. Identify cities with total order value above 10,000.

```
SELECT m.city, SUM(b.price * o.quantity) AS total_value
FROM orders o
JOIN members m ON o.member_id = m.member_id
JOIN books b ON o.book_id = b.book_id
GROUP BY m.city
HAVING total_value > 10000;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
	city	total_value	
▶	Kochi	15630.50	
	Thrissur	17563.54	
	Kozhikode	12100.77	
	Ernakulam	15181.49	
	Alappuzha	11125.43	

#12. Count total number of orders per city

```
SELECT m.city, COUNT(*) AS total_orders
FROM orders o
JOIN members m ON o.member_id = m.member_id
GROUP BY m.city;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
	city	total_orders	
▶	Kochi	159	
	Thrissur	129	
	Kozhikode	124	
	Ernakulam	169	
	Alappuzha	115	
	Trivandrum	56	
	Kottayam	48	

#13. Find the most popular genre by number of orders

```
SELECT b.genre, COUNT(*) AS order_count
FROM orders o
JOIN books b ON o.book_id = b.book_id
GROUP BY b.genre
ORDER BY order_count DESC
LIMIT 1;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
	genre	order_count		
▶	Fiction	134		

3.Advanced

#14.Find members whose total spending is above average member spend.

```
SELECT m.name, SUM(b.price*o.quantity) AS total_spent
FROM orders o
JOIN members m ON o.member_id = m.member_id
JOIN books b ON o.book_id = b.book_id
GROUP BY m.name
HAVING total_spent > (
    SELECT AVG(spend) FROM (
        SELECT SUM(b2.price*o2.quantity) AS spend
        FROM orders o2 JOIN books b2 ON o2.book_id=b2.book_id
        GROUP BY o2.member_id
    ) AS sub
);
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
name	total_spent		
▶ Sneha Menon	1089.31		
Anu Jose	1445.62		
Manoj Kumar	2639.17		
Aleena Varma	1097.56		
Aleena Nambiar	1585.72		
Ajay Menon	3900.70		
Arun Nair	3846.31		
Vishnu Menon	2313.26		
Vishnu Iyer	1267.54		

#15. Rank genres by total revenue using RANK().

```
SELECT b.genre, SUM(b.price*o.quantity) AS revenue,
RANK() OVER (ORDER BY SUM(b.price*o.quantity) DESC) AS genre_rank
FROM orders o JOIN books b ON o.book_id = b.book_id
GROUP BY b.genre;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
genre	revenue	genre_rank	
▶ Computers	19016.15	1	
Business & Economics	13110.13	2	
Literary Criticism	7429.62	3	
Language Arts & Disciplines	6377.72	4	
Philosophy	5543.05	5	
History	5087.80	6	
Psychology	3780.10	7	
Technology & Engineering	3269.94	8	
Social Science	3194.48	9	

#16. Analyze monthly sales trend using a CTE.

```
WITH monthly AS (  
    SELECT DATE_FORMAT(o.order_date, '%Y-%m') AS month, SUM(b.price*o.quantity) AS revenue  
    FROM orders o JOIN books b ON o.book_id = b.book_id  
    GROUP BY month  
)  
SELECT * FROM monthly ORDER BY month;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
	month	revenue	
▶	2024-01	7956.90	
	2024-02	6184.08	
	2024-03	12285.03	
	2024-04	3860.45	
	2024-05	6553.99	
	2024-06	5169.22	
	2024-07	7269.74	
	2024-08	11195.01	
	2024-09	5122.20	

#17. Create a reusable view summarizing total orders and revenue by city;

```
CREATE VIEW city_summary AS  
SELECT m.city, COUNT(*) AS total_orders, SUM(b.price*o.quantity) AS total_revenue  
FROM orders o  
JOIN members m ON o.member_id = m.member_id  
JOIN books b ON o.book_id = b.book_id  
GROUP BY m.city;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
	city	total_orders	total_revenue
▶	Kochi	159	15630.50
	Thrissur	129	17563.54
	Kozhikode	124	12100.77
	Ernakulam	169	15181.49
	Alappuzha	115	11125.43
	Trivandrum	56	6241.48
	Kottayam	48	8696.83

#18. Rank members by number of orders placed (window function)

```
SELECT m.name, COUNT(*) AS order_count,  
RANK() OVER (ORDER BY COUNT(*) DESC) AS member_rank  
FROM orders o  
JOIN members m ON o.member_id = m.member_id  
GROUP BY m.name;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
	name	order_count	member_rank
▶	Ajay Menon	29	1
	Vishnu Menon	26	2
	Arun Nair	24	3
	Lakshmi Jose	24	3
	Meera Jose	22	5
	Anu Iyer	20	6
	Manoj Kumar	20	6
	Aleena Nambiar	17	8

46 #19. Find books that have never been ordered (subquery)

47 • **SELECT * FROM** books

48 **WHERE** book_id **NOT IN** (SELECT DISTINCT book_id FROM orders);

49

Result Grid



Filter Rows:

Edit:




Export/Import:



Wrap Cell Content:


	book_id	title	author	genre	price
▶	3	Hype	Jon Helgason	Literary Criticism	31.99
	11	Christian Books	Ivan King	Religion	9.99
	13	Christian Suspense	Ivan King	Young Adult Fiction	9.99
	16	Young Adults	Ivan King	Young Adult Fiction	9.99
	25	In the Dead of the Night	Terry Spear	Fiction	4.99
	28	The Little Princess of Tower Hill	L. T. Meade	Unknown	8.30
	33	The Stuff of Science Fiction	Gary Westfahl	Literary Criticism	29.99
	41	The War of The Worlds - A Science Fiction Class...	H. G. Wells	Fiction	0.99

#20. Calculate the return/cancellation rate per genre

```
SELECT b.genre,  
ROUND(SUM(CASE WHEN o.order_status IN  
( 'Cancelled', 'Returned' ) THEN 1 ELSE 0 END) / COUNT(*) * 100, 2)  
AS issue_rate_pct  
FROM orders o  
JOIN books b ON o.book_id = b.book_id  
GROUP BY b.genre;
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	genre	issue_rate_pct			
▶	Science	43.75			
	Law	0.00			
	Art	33.33			
	Fiction	40.30			
	Biography & Autobiography	38.10			
	Literary Criticism	27.42			
	Philosophy	38.46			
	Business & Economics	37.25			